SPINE · NEET-SS / FET / INI-SS

Scoliosis and deformity

1. Definitions and classification

- **Scoliosis:** lateral curvature **over 10 degrees (Cobb angle)** with vertebral rotation; a three-dimensional deformity (coronal curve, axial rotation, sagittal **hypokyphosis** in the thoracic spine in idiopathic scoliosis).
- **Types: idiopathic** (about 80%; infantile 0-3, juvenile 4-10, **adolescent 10-skeletal maturity**), **congenital** (vertebral malformation), **neuromuscular** (CP, DMD, SMA, myelomeningocele, Friedreich), **syndromic** (Marfan, NF1, Ehlers-Danlos), **degenerative (de novo adult)**, others (tumour: osteoid osteoma; painful; leg-length discrepancy).
- **Early-onset scoliosis (EOS):** onset **before 10 years** of any cause; risks **thoracic insufficiency syndrome** (impaired lung growth: alveolar development mostly before about 8 years).

2. Adolescent idiopathic scoliosis (AIS)

Epidemiology and features

- **Prevalence about 2-3%** (curves over 10 degrees); **girls:boys about 1:1 for small curves but about 8-10:1 for curves needing treatment**. Most common pattern: **right thoracic** curve (left thoracic curves are atypical: MRI to exclude **syrinx, Chiari malformation, tethered cord**).
- **Painless** (significant pain, neurological signs, **left thoracic curve**, rapid progression, **abnormal abdominal reflexes**, cavus foot, **male with a severe curve**, early onset: get an **MRI of the whole neural axis**).

- **Adams forward bending test: rib hump** (rotation); **scoliometer reading 5-7 degrees or more** = refer for radiograph (screening threshold about 7 degrees corresponds to about a 20-degree Cobb angle).

Radiographic assessment

- **Standing PA whole spine** (low-dose EOS imaging where available). **Cobb angle:** between the end plates of the **most tilted end vertebrae** (measurement error about ±5 degrees).
- **Risser sign:** ossification of the iliac apophysis (lateral to medial): **0 none, 1 = 25%, 2 = 50%, 3 = 75%, 4 = 100% (not fused), 5 = fused to the ilium**. The **peak height velocity** occurs **before Risser 1** (at about **Risser 0 with open triradiate cartilage**), so Risser underestimates remaining growth. **Sanders skeletal maturity staging** (hand radiograph: digital epiphyses; **Sanders 3 = peak growth**) is a better predictor. **Menarche** usually comes about 1 year after peak height velocity.
- **Lenke classification** (for surgery planning; 6 curve types + lumbar modifier A/B/C + sagittal thoracic modifier −/N/+):

Lenke type	Description	Structural curves
1	**Main thoracic** (commonest)	MT
2	Double thoracic	PT + MT
3	Double major	MT + TL/L
4	Triple major	PT + MT + TL/L
5	Thoracolumbar/lumbar	TL/L
6	Thoracolumbar/lumbar-main thoracic	TL/L (major) + MT

  - A curve is **structural** if it remains **25 degrees or more on side-bending** (or has kyphosis of 20 degrees or more in the PT or TL region). Lumbar modifier: **A** (CSVL between the pedicles of the

lumbar apex), **B** (touches apical body/pedicle), **C** (CSVL entirely medial to the apical vertebra).

- **King-Moe classification** (older; types I-V).
- **Other measures:** apical vertebral rotation (**Nash-Moe**), trunk shift, coronal and sagittal balance, **shoulder balance**, rib-vertebra angle difference (**RVAD; Mehta**: in infantile scoliosis, **RVAD over 20 degrees** and **phase 2 rib head overlap** predict progression).

Natural history (progression)

- **Risk of progression rises with: curve magnitude**, **skeletal immaturity** (Risser 0-1, premenarchal, open triradiate, Sanders under 4), **female sex**, **thoracic and double curves**.
- **After maturity:** thoracic curves **over 50 degrees** progress about **1 degree per year**; curves under 30 degrees rarely progress. **Thoracic curves over about 70-100 degrees** reduce pulmonary function (restrictive); **over about 100 degrees** risk cor pulmonale. Lumbar curves cause back pain and degenerative changes later.

Treatment (SRS principles)

Curve (skeletally immature)	Treatment
Under 20-25 degrees	**Observation** (radiographs every 4-6 months)
25-40 (45) degrees, Risser 0-2	**Bracing**
Over 45-50 degrees	**Surgery** (posterior spinal fusion)

- **BrAIST trial (Weinstein, NEJM 2013):** bracing **significantly reduced progression to surgical threshold (50 degrees)** (success about 72% with bracing vs 48% with observation) with a **dose-response**: **wear over about 13 hours/day** gave about 90% success. **Full-time TLSO (Boston)** about 18-23 hours; night-time brace (Charleston, Providence) for single lumbar/TL curves. Bracing **halts progression;**

it does not correct the curve. Bracing is ineffective for curves over 45 degrees or after maturity.

- **Scoliosis-specific exercises (Schroth):** some evidence as adjunct.
- **Surgery: posterior spinal instrumentation and fusion with pedicle screws** (all-pedicle-screw constructs give the best three-dimensional correction), selective fusion of structural curves (**Lenke 1: fuse the main thoracic curve only**, lumbar curve corrects spontaneously when lumbar modifier A/B; selective thoracic fusion in some C modifiers). **Anterior fusion** (rarely now) for thoracolumbar/lumbar curves to save levels. **Vertebral body tethering** (growth modulation, in skeletally immature patients; emerging).
- **Neuromonitoring (SSEPs and MEPs; Stagnara wake-up test as backup): MEPs detect motor tract compromise sooner**. On signal loss: raise MAP, check anaesthetic/temperature, release correction, check screws, steroids debated, wake-up test.
- **Complications: neurological deficit (about 0.3-1%)**, infection (late infection with *Cutibacterium acnes*), **pseudarthrosis**, **crankshaft phenomenon** (posterior fusion alone in a very immature spine (Risser 0, open triradiate): anterior growth continues, increasing rotation and curve; prevent with anterior release/fusion or pedicle screws), **adding-on** (curve extends distally), **flatback** (loss of lumbar lordosis with distraction instrumentation: Harrington), **SMA syndrome (superior mesenteric artery syndrome: vomiting after correction from compression of the duodenum)**, **shoulder imbalance**, blood loss (tranexamic acid reduces it), **proximal junctional kyphosis**.

3. Early-onset and infantile scoliosis

- **Infantile idiopathic scoliosis: boys**, **left thoracic** curves, associated plagiocephaly, hip dysplasia; **most resolve**. **Mehta: RVAD over 20**

degrees and phase 2 rib head = progressive. **Mehta casting** (derotational serial casting under GA) can cure early curves.

- **Growth-friendly surgery (EOS): traditional growing rods** (lengthening every 6 months; many operations, autofusion), **magnetically controlled growing rods (MCGR)** (non-invasive lengthening), **VEPTR** (vertical expandable prosthetic titanium rib; for thoracic insufficiency, congenital fused ribs), **Shilla** (guided growth). Definitive fusion at maturity. **Law of diminishing returns** with repeated lengthening.
- **Juvenile idiopathic scoliosis:** high progression risk (about 70% need treatment); MRI common (higher rate of neural axis abnormality).

4. Congenital scoliosis

- **Failure of formation** (**hemivertebra**: fully segmented, semi-segmented, non-segmented, incarcerated; wedge vertebra), **failure of segmentation** (**unilateral unsegmented bar**, block vertebra), **mixed**.
- **Worst prognosis: unilateral unsegmented bar with contralateral fully segmented hemivertebra** (progression about **5-10 degrees per year**; **early surgery**). **Block vertebra** has the best prognosis.
- **Associated anomalies (screen everyone): renal (about 20-30%: renal ultrasound)**, **cardiac (about 10-25%: echocardiography)**, **intraspinal (about 20-40%: MRI: diastematomyelia, tethered cord, syrinx)**, **VACTERL**, **Klippel-Feil** (cervical fusion, low hairline, short neck: Sprengel deformity, renal and hearing anomalies), Goldenhar, Alagille.
- **Bracing does not work** (rigid bony curves). **Surgery: hemivertebra excision with short fusion** (posterior-only, at a young age, ideally under about 5), in situ fusion (convex growth arrest/hemi-epiphysiodesis), growth-friendly systems, osteotomies for established curves.

- **Congenital kyphosis** (anterior failure of formation, type I) has a high risk of **paraplegia**: early posterior fusion. Type II (failure of segmentation) is less dangerous.

5. Neuromuscular and syndromic scoliosis

- **Neuromuscular:** long **C-shaped thoracolumbar curves** with **pelvic obliquity**, early onset, progression **after maturity**, poor response to bracing (brace for sitting only). Fuse when sitting balance worsens or curve over about 40-50 degrees; **fusion to the pelvis** if pelvic obliquity (iliac or **S2-alar-iliac screws**); pre-operative pulmonary and nutritional optimisation. **DMD: fuse early (20-30 degrees, FVC over about 30-35%).** Higher complication rates (infection, blood loss, pulmonary).
- **NF1: dystrophic curves** (short, sharp, **rib penciling**, scalloping, dural ectasia, enlarged foramina): **progress rapidly**, need early **anterior and posterior fusion**; non-dystrophic behave like idiopathic. **Modulation** of non-dystrophic into dystrophic curves.
- **Marfan:** larger, more rigid curves, dural ectasia (pedicle screw problems), **higher pseudarthrosis and complication rates**.

6. Kyphosis

- **Normal thoracic kyphosis about 20-45 degrees** (T5-T12); **lumbar lordosis about 40-60 degrees**.
- **Postural kyphosis:** flexible, corrects on prone extension.
- **Scheuermann kyphosis: 3 or more adjacent vertebrae with 5 degrees or more of anterior wedging** (Sorensen), endplate irregularity, **Schmorl nodes**, narrowed discs; **rigid** thoracic kyphosis **over 45 degrees** (does not correct on hyperextension); adolescent males; hamstring tightness, compensatory lumbar hyperlordosis, **spondylolysis association**. **Bracing (Milwaukee/modified)** for curves about 50-75 degrees in skeletally immature patients; **posterior fusion** (with posterior column osteotomies) for **over 75**

degrees, pain or deformity; fuse from the proximal end vertebra to the **sagittal stable vertebra** (to avoid **distal junctional kyphosis**).

- **Congenital kyphosis**, **post-traumatic**, **post-laminectomy** (children, cervical), **tuberculous** (gibbus), **ankylosing spondylitis** (chin-on-chest; **pedicle subtraction osteotomy** at L3 or cervicothoracic osteotomy), **osteoporotic**.

7. Adult deformity (brief; see spinopelvic notes)

- **Adult idiopathic** (untreated AIS) vs **de novo degenerative scoliosis** (lumbar, older patients, asymmetric disc degeneration, rotatory subluxation, stenosis).
- **SRS-Schwab classification:** coronal curve type (T, L, D double, N none) + sagittal modifiers: **PI-LL mismatch (0 under 10 degrees, + 10-20, ++ over 20)**, **pelvic tilt (0 under 20, + 20-30, ++ over 30)**, **SVA (0 under 4 cm, + 4-9.5, ++ over 9.5 cm)**. **Sagittal imbalance correlates most with disability.**
- **Osteotomies: Smith-Petersen/Ponte (posterior column; about 10 degrees per level, needs a mobile anterior disc)**, **pedicle subtraction osteotomy (PSO; about 30-35 degrees, three-column, closing wedge through the vertebra)**, **vertebral column resection (VCR; over 40 degrees, rigid multiplanar deformity; highest neurological risk)**.
- **Complications: proximal junctional kyphosis/failure**, pseudarthrosis (rod fracture at PSO levels), neurological deficit, high blood loss, medical complications.

> Exam pearl: **AIS: girls, right thoracic; left thoracic, pain, neurology or abnormal abdominal reflexes = MRI. Peak growth before Risser 1 (Sanders 3). Treatment: observe under 25, brace 25-40/45 if immature, fuse over 45-50. BrAIST: bracing works, over 13 h/day.**

Exam pearl: **Congenital scoliosis: unilateral unsegmented bar + contralateral hemivertebra = worst; screen kidneys (ultrasound), heart (echo) and spinal cord (MRI). Bracing does not help. Crankshaft: posterior fusion alone in a very immature spine.**

Exam pearl: **Scheuermann: 3 adjacent vertebrae with 5 degrees or more wedging, rigid kyphosis over 45 degrees; fuse to the sagittal stable vertebra. Ponte about 10 degrees/level, PSO about 30 degrees, VCR over 40 degrees.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.